Mizba

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Education

B.Tech Artificial Intelligence & Data Science 2023–2027
B.S. Abdur Rahman Crescent Institute of Science and Technology, Chennai CGPA: 9.20

Higher Secondary Education (Grade 12) 2023
Kerala State Board 98%

Secondary Education (Grade 10) 2021
CBSE 93%

Projects

Emergency Incident Fusion System (AI-01) *Prismatic 2k26 | Google Solution Challenge 2026*
React / Vite / Express / TypeScript / Supabase / pgvector / Sarvam AI / Gemini 2.5 Flash

- Built a multilingual emergency report fusion platform that processes voice and text incident reports in regional Indian languages using Sarvam AI's Saaras v3 STT and Bulbul v3 TTS pipelines
- Designed a five-metric Composite Fusion Score (CFS ≥ 0.72) combining semantic similarity (30%), geospatial proximity (25%), temporal overlap (20%), category match (15%), and entity overlap (10%)
- Implemented semantic deduplication of overlapping reports using pgvector with OpenAI `text-embedding-3-small`, reducing duplicate alerts by over 60%
- Integrated Gemini 2.5 Flash for vision-based media analysis of uploaded incident images and video frames
- Presented at Prismatic 2k26 National Symposium and shortlisted for Google Solution Challenge 2026

Student Academic Performance Predictor
Python / scikit-learn / XGBoost / Pandas / Streamlit / Seaborn

- Designed an ML pipeline to predict student at-risk status using attendance, assessment scores, and engagement signals from 1,200+ anonymised student records
- Applied feature engineering, outlier handling, and SMOTE oversampling to address class imbalance in the training set
- Evaluated Random Forest, XGBoost, and Logistic Regression; XGBoost achieved best accuracy of 88.4% and F1-score of 0.87
- Deployed an interactive Streamlit dashboard for faculty to visualise semester-wise performance trends and generate at-risk reports

Tamil–English Fake News Detection System
Python / HuggingFace Transformers / FastAPI / React / PostgreSQL

- Built a bilingual misinformation classifier for Tamil and English news articles by fine-tuning `ai4bharat/indic-bert` on a curated 10,000-sample dataset
- Achieved an F1-score of 0.91 on the held-out test set; deployed a FastAPI inference endpoint consumed by a React frontend
- Applied LIME-based post-hoc explainability to surface token-level evidence for each classification decision
- Automated dataset curation pipeline using web scraping and Google Fact Check API for label generation

Technical Skills

Programming	Python, SQL, JavaScript, C
ML / DL Frameworks	TensorFlow, PyTorch, scikit-learn, XGBoost, HuggingFace Transformers
Data Tools	Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook
Web & Backend	React, Vite, FastAPI, Express, TypeScript
Databases	PostgreSQL, Supabase, pgvector, Firebase
Tools & Platforms	Git, GitHub, Google Colab, Streamlit, VS Code, Figma

Achievements

- Shortlisted for **Google Solution Challenge 2026** with the AI-01 Emergency Incident Fusion System
- Presented research at **Prismatic 2k26**, National Level Technical Symposium, Easwari Engineering College
- Secured **98%** in Higher Secondary (Kerala State Board) – top performer in the school
- Consistent academic record with **CGPA 9.20** across five semesters of B.Tech

Certifications

- Machine Learning Specialization – Andrew Ng / Coursera

- Deep Learning with TensorFlow – IBM / Coursera
- Joy of Computing using Python – NPTEL